LONGEVITY 17

Somatropin (growth hormone)

Recombinant human growth hormone, a 191-amino-acid protein given by injection under the skin



CLASS	STUDIED FOR	HUMAN EVIDENCE	US STATUS
Hormone	GH deficiency, body composition	Large trials; FDA approved	FDA approved (Genotropin) · not for aging

STUDIED FOR

- **GH deficiency** — HUMAN
- **Child growth disorders** — HUMAN
- **HIV wasting** — HUMAN
- **Lean mass and fat** — HUMAN

Aging is not an approved use. US federal law makes it a crime to distribute human growth hormone for any use FDA has not authorized, including anti-aging.

HOW IT WORKS IN THE BODY

Somatropin ▶ IGF-I levels ▶ Growth plates ▶ Lean mass, growth

- **Equivalent to pituitary growth hormone** in lab and clinical tests (label)
- **Stimulates linear growth** in children; normalizes IGF-I in GH deficiency (label)
- **In adults with GH deficiency,** less fat mass and more lean mass (label)
- **In aging,** a benefit is not established; strength did not clearly rise

EVIDENCE & WHAT'S KNOWN

EVIDENCE

FDA APPROVED
LEVEL 5 OF 5

- Approved for growth disorders since the 1980s
- Older adults: 18 small RCTs reviewed (2007)
- No trial showed slower aging

NOT YET ESTABLISHED

- Effect on aging in people
- Strength gain in older adults
- Fertility in people (no data)

QUICK PRACTICAL INFO

HALF-LIFE	ROUTES STUDIED
3.0 hours under the skin (GHD adults); 0.4 hours IV	Injection under the skin
PHARMACOKINETICS	**SOLD AS**
Protein breakdown in liver and kidneys (label)	Prescription injection (Genotropin and other brands)
MONITORED IN TRIALS	**SPORT (WADA)**
IGF-I levels, blood sugar, thyroid, cortisol, tumors	Prohibited, class S2

REGULATORY

FDA	Approved for growth disorders, GHD, HIV wasting
FDA	Distributing for unapproved uses is a crime
WADA	Prohibited at all times (S2.2.3)

REFERENCES

1. FDA, Genotropin prescribing information, Aug 2026
2. Ann Intern Med, growth hormone in healthy elderly: systematic review, Jan 2007
3. WADA, 2026 Prohibited List

Adonis Men's Health · checked Nov 2026. Research and regulator records only. Not medical advice.

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